

Termodinámica

0.1 Entropía

$$\begin{aligned}S &= S(U, V, N) \\dS &= \frac{1}{T}dU + \frac{P}{T}dV - \frac{\mu}{T}dN \\dU &= TdS - pdV + \mu dN \\S &= \frac{1}{T}U + \frac{P}{T}V - \frac{\mu}{T}N \text{ (Solo sist ext)}\end{aligned}$$

0.2 Energía de Hemholtz

$$\begin{aligned}F &= U - TS \\dF &= -SdT - pdV + \mu dN\end{aligned}$$

0.3 Energía de Gibbs

$$\begin{aligned}G &= U - TS + pV \\dG &= -SdT + vdp + \mu dN\end{aligned}$$

0.4 Hentalpía

$$\begin{aligned}H &= U + pV \\dH &= TdS + vdp + \mu dN\end{aligned}$$

0.5 Gran Canónico

$$\begin{aligned}\Xi &= U - TS - \mu N \\d\Xi &= -SdT - pdV - Nd\mu\end{aligned}$$

Ecuaciones de Maxwell y Cv

$$\begin{aligned}+\left(\frac{\partial T}{\partial V}\right)_S &= -\left(\frac{\partial p}{\partial S}\right)_V = \frac{\partial^2 U}{\partial S \partial V} \\+\left(\frac{\partial T}{\partial p}\right)_S &= +\left(\frac{\partial V}{\partial S}\right)_p = \frac{\partial^2 H}{\partial S \partial p} \\+\left(\frac{\partial S}{\partial V}\right)_T &= +\left(\frac{\partial p}{\partial T}\right)_V = \frac{\partial^2 F}{\partial T \partial V} \\-\left(\frac{\partial S}{\partial p}\right)_T &= +\left(\frac{\partial V}{\partial T}\right)_p = \frac{\partial^2 G}{\partial T \partial p} \\C_V &= T\left(\frac{\partial S}{\partial T}\right)_V\end{aligned}$$

Combinatoria

0.6 Stirling

$$\begin{aligned}\log N! &\approx N \log N - N \\ \log \binom{N}{xN} &\approx -N \left[x \log x + (1-x) \log(1-x) \right]\end{aligned}$$

0.7 Inclusión - Exclusión

Suele servir cuando quiero imponer condiciones restrictivas

$$\begin{aligned}P(n, m) &= \sum_{i=1}^m N(A_i) - \sum_{i,j} ' N(A_i \cap A_j) + \\ &\quad \sum_{i,j,k} ' N(A_i \cap A_j \cap A_k) - \dots\end{aligned}$$

0.8 Bayes

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

De estas se deduce:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

0.9 Aprox término máximo

$$\lim_{N \rightarrow \infty} \frac{1}{N} \log Z = g(x^*) \rightarrow \log \sigma \approx Ng(x) + o(N)$$

Ensambls

0.10 Microcanónico

Fijos: E, V, N

$\Omega_m \rightarrow$ (Problema de combinatoria)

$$\begin{aligned}P_m &= \frac{1}{\Omega_m} \\S &= k \log(\Omega_m)\end{aligned}$$

$$\Omega(E, V, N) = \int_{E-\Delta E \leq H \leq E} \frac{d^{3N} q d^{3N} p}{h^{3N}}$$

0.11 Canónico

Fijos: V, N

$$\begin{aligned}Z &= \sum_{V, N} e^{-\beta E_m} \\ &= \sum_E \Omega(E) e^{-\beta E} \\ &= \int \frac{d^{3N} q d^{3N} p}{h^{3N}} e^{-\beta \sum_j H(j)} \\ F &= -kT \log Z \\ \langle E \rangle &= -\frac{\partial \log Z}{\partial \beta} \\ Var(E) &= -\frac{\partial \langle E \rangle}{\partial \beta} \\ S &\approx k \log \sigma(n) - y \log x\end{aligned}$$

0.12 Gran canónico

$$\begin{aligned}\mathcal{Z} &= \sum_V z^{N_m} e^{-\beta E_m} \\ z &= e^{\beta \mu} \\ \Xi &= -kT \log \mathcal{Z} \\ \langle E \rangle &= -\frac{\partial \log \mathcal{Z}}{\partial \beta} \\ \langle N \rangle &= z \frac{\partial \log \mathcal{Z}}{\partial z}\end{aligned}$$

Gas ideal

Recordar el factor de corrección de Boltzman!

$$S = kN \left[\log \left(\frac{V}{N\lambda^3} \right) + \frac{5}{2} \right] \rightarrow \lambda = \frac{h}{\sqrt{2\pi mkT}}$$

$$E = \frac{3}{2} NkT$$

$$pV = NkT$$

$$Z = (Z_1)^{3N}$$

$$Z_1 = \frac{V}{\lambda^3}$$

$$\mathcal{Z} = e^{Z_1 z}$$

$$z = p\beta\lambda^3$$

$$\langle N \rangle = zZ_1 = \frac{zV}{\lambda^3}$$

Utilidades

0.13 Cosas de combinatoria

$$\binom{N}{m} = \binom{N-1}{m} + \binom{N-1}{m-1}$$

$$\sum_{i=1}^N i = \frac{n(n+1)}{2}$$

$$\sum_{i=0}^n r^i = \frac{1-r^{n+1}}{1-r}$$

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r} \quad |r| < 1$$

$$(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

$$(a-b)^n = \sum_{k=0}^n (-1)^k \binom{n}{k} a^{n-k} b^k$$

Hiper-cosas

$$\text{Hiper-esfera:} \quad V_D(R) = \frac{\pi^{D/2} R^D}{(D/2)!}$$

$$\text{Hiper-pirámide:} \quad V_n = \frac{1}{n} A_{n-1} h$$

$$\text{Hiper-elipsoide:} \quad V_n = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)} \prod_{i=1}^n a_i$$